Convexity of Hadamard products in the class Ω

Principia Math in Progress

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Task 1. Resolve MathDB 343776

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and let $\mathcal{A}$ consist of analytic functions $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ on $\mathbb{D}$. Define

$$\Omega = \{f \in \mathcal{A} : |zf'(z) - f(z)| < 1/2 \text{ for every } z \in \mathbb{D}\}.$$

The unnumbered conjecture on page 19 of [1], retained as the open Problem on published page 1424, asks whether $f_1 * f_2$ is convex for every $f_1, f_2 \in \Omega$. Here $*$ denotes coefficientwise multiplication.

Theorem 1. *If $f_1(z) = z + \sum_{n\geq 2} a_n z^n$ and $f_2(z) = z + \sum_{n\geq 2} b_n z^n$ belong to Ω, then*

$$\sum_{n=2}^{\infty} n^2 |a_n b_n| \leq 1. \tag{1}$$

Their Hadamard product is univalent and maps $\mathbb{D}$ onto a convex domain.

Proof. For $0 < \rho < 1$, Parseval's identity applied to $zf_1'(z) - f_1(z)$ gives

$$\sum_{n=2}^{\infty} (n-1)^2 |a_n|^2 \rho^{2n} = \frac{1}{2\pi} \int_0^{2\pi} |\rho e^{it} f_1'(\rho e^{it}) - f_1(\rho e^{it})|^2 \, dt \leq \frac{1}{4}.$$

Monotone convergence as $\rho \uparrow 1$ therefore gives

$$E_a := \sum_{n\geq 2} (n-1)^2 |a_n|^2 \leq \frac{1}{4}, \qquad E_b := \sum_{n\geq 2} (n-1)^2 |b_n|^2 \leq \frac{1}{4}.$$

Since $n^2 \leq 4(n-1)^2$ for every integer $n \geq 2$, Cauchy–Schwarz implies

$$\sum_{n\geq 2} n^2 |a_n b_n| \leq 4 \sum_{n\geq 2} (n-1)^2 |a_n||b_n| \leq 4\sqrt{E_a E_b} \leq 1.$$

In particular $h(z) := z + \sum_{n\geq 2} a_n b_n z^n$ is analytic on $\mathbb{D}$. To check convexity directly, put $c_n = a_n b_n$ and, for $0 < r < 1$, define

$$A(r) = \sum_{n\geq 2} n|c_n| r^{n-1}, \qquad B(r) = \sum_{n\geq 2} n(n-1)|c_n| r^{n-1}.$$

Equation (1) yields

$$A(r) + B(r) = \sum_{n\geq 2} n^2 |c_n| r^{n-1} \leq r < 1.$$

For $|z| = r$ we consequently have

$$|h'(z)| \geq 1 - A(r) > 0, \qquad |zh''(z)| \leq B(r) < 1 - A(r) \leq |h'(z)|.$$

It follows that $\mathrm{Re}(1 + zh''(z)/h'(z)) > 0$ throughout $\mathbb{D}$; the value at zero is 1. The standard analytic characterization of convex functions now proves the assertion. This characterization is also the definition of $\mathcal{C}$ used on published page 1410 of [1]. □

Sharpness and scope. The pair $f_1 = f_2 = z + z^2/2$ belongs to Ω and gives $h = z + z^2/4$, attaining equality in (1). Moreover

$$1 + \frac{zh''(z)}{h'(z)} = \frac{1+z}{1+z/2}, \qquad \lim_{r\uparrow 1} \frac{1-r}{1-r/2} = 0.$$

Thus a positive uniform order of convexity cannot be asserted for all such products. The argument allows arbitrary complex coefficients and does not assume a coefficientwise sufficient condition on either factor.

Task 2. Quantify stability near equality. Derive explicit bounds on the weighted square sums of the higher coefficients when the left side of (1) exceeds $1 - \varepsilon$, and determine the optimal dependence on ε.

Task 3. Determine the optimal order for unequal bounds. For $\Omega_\alpha = \{f \in \mathcal{A} : |zf' - f| < \alpha\}$, determine the best uniform order of convexity of $f * g$ with $f \in \Omega_\alpha$ and $g \in \Omega_\beta$, including the extremal pairs in the convex range.

Task 4. Characterize convolution multipliers. Characterize all normalized analytic functions g for which $f * g$ is convex for every $f \in \Omega$, and compare this class with Ω.

Task 5. Determine stronger geometric radii. Find the sharp radius on which every product of two members of Ω is uniformly convex, and extend the question to products of three or more members.

References

[1] A. Swaminathan and L. A. Wani, *Sufficient Conditions and Radius Problems for a Starlike Class Involving a Differential Inequality*, Bull. Korean Math. Soc. 57 (2020), 1409–1426. doi:10.4134/BKMS.b191074. Preprint arXiv:1909.04471.